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Business process management for Industry 4.0 – Three application cases in the DFKI-Smart-Lego-Factory

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Abstract: The advent of Industry 4.0 is expected to dramatically change the manufacturing industry as we know it today. Highly standardized, rigid manufacturing processes need to become self-organizing and decentralized. This flexibility leads to new challenges to the management of smart factories in general and production planning and control in particular. In this contribution, we illustrate how established techniques from Business Process Management (BPM) hold great potential to conquer challenges in Industry 4.0. Therefore, we show three application cases based on the DFKI-Smart-Lego-Factory, a fully automated “smart factory” built out of LEGO® bricks, which demonstrates the potentials of BPM methodology for Industry 4.0 in an innovative, yet easily accessible way. For each application case (model-based management, process mining, prediction of manufacturing processes) in a smart factory, we describe the specific challenges of Industry 4.0, how BPM can be used to address these challenges, and, their realization within the DFKI-Smart-Lego-Factory.

Keywords: Industry 4.0, Business Process Management, Smart Factories, Process Mining, Process Prediction

ACM CCS: Applied computing → Enterprise Computing → Business Process Management

1 Introduction

The advent of Industry 4.0 is expected to dramatically change the manufacturing industry. While the technical potentials of the industrial digitalization appear endless, they also pose a number of challenges to stakeholders involved in managing processes and production plants.

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Manufacturing processes in particular need to change in a completely unknown way, enabling an automated production of low product quantities with a high variant diversity. This flexibility leads to new requirements to factory management in general and production planning and control in particular. Planning and modeling both the product and its production processes in an integrated way with sufficient degrees of freedom is crucial to define how systems communicate and share information for value creation. This means that the drastic changes Industry 4.0 will have on the manufacturing industry also significantly impact the process management in a smart factory. Not only do they call for an intelligent production as such, but rather the intelligent design of the entire value chain.

Applying established techniques from Business Process Management (BPM) to Industry 4.0 holds great potential for conquering these challenges. BPM provides stakeholders with sufficient means to manage intelligent manufacturing processes and smart factories efficiently and effectively. Considering manufacturing from a process perspective facilitates bridging the abstraction gap between individual sensor data and high-level process steps as well as cross-divisional and cross-organizational integration. Meanwhile, the new technological potential gives rise to innovative BPM methods, e. g., by collecting more process execution data on a higher level of detail.

In this paper, we illustrate the potentials of BPM for Industry 4.0 with three application cases. They are based on the DFKI-Smart-Lego-Factory, a fully automated “smart factory” out of LEGO® bricks, which presents BPM for Industry 4.0 in an innovative, yet easily accessible way. After presenting foundations of BPM and Industry 4.0 in Sect. 2, we describe the DFKI-Smart-Lego-Factory in Sect. 3 and describe the cases in sections 4 to 6, before the paper is concluded in Sect. 7.

2 Foundations

2.1 The BPM lifecycle

BPM can be considered as a continuous evolution with several stages, often conceptualized as a lifecycle shown in Fig. 1 [5]. It consists of six stages:

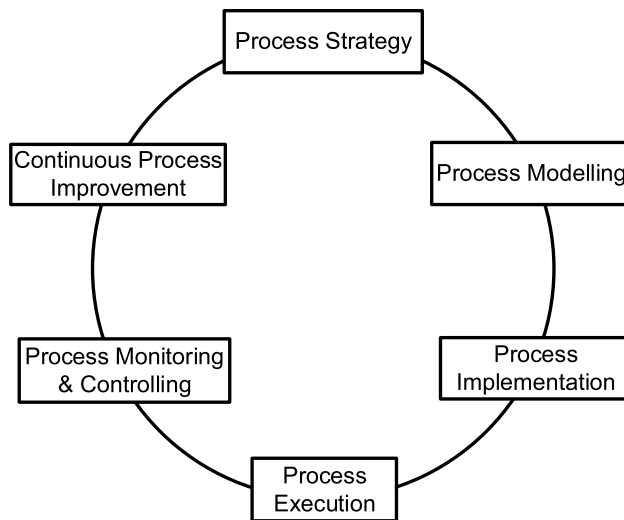


Figure 1: The BPM lifecycle [5].

1. **Process Strategy:** Role and objective of a process are determined by the business model of the respective company. So, before (re-)engineering a process, stakeholders need to be aware of its positioning within the company strategy in general and the business model in particular. This includes identifying the contribution a process makes towards the business model and hence, defining its strategic goals (e. g., by means of KPIs) and relevance to the company as a whole.
2. **Process Modeling:** In order to define the process in detail, it has to be designed and documented by means of process modeling methods. All aspects of the process, including the individual steps, but also, e. g., the related data objects or organizational resources, can be included in a (semi-)formal process model using an appropriate modeling language and the required levels of detail and abstraction.
3. **Process Implementation:** After processes have been defined and modeled, they have to be implemented within the organization. Process implementation entails the establishment of all structures that are necessary to realize the process as defined by the models and further documentations, including, e. g., IT systems, but also organizational structures.
4. **Process Execution:** Finally, the implemented structures are used to execute the defined process according to its strategic goal and purpose. Ideally, during the course of process execution, execution data (such as process logs) is collected, which documents the process execution and enables data-centric monitoring and controlling in the next step.

5. **Process Monitoring & Controlling:** Each running process within an organization is monitored and controlled during its execution. By examining the results of the previous stages, i. e., the process models, the established execution structures, and the collected execution data, stakeholders can identify potential shortcomings, which allow to further optimize the process as a whole.
6. **Continuous Process Improvement:** Based on the process monitoring results, a running process needs to be continually adapted and improved, to adhere to changing external requirements, but also to reach its defined goals in an optimized, more resource-efficient way. To realize optimization potentials, the process needs to pass through the lifecycle again, identifying the impact each change has on the individual process management stages.

2.2 Industry 4.0

The omnipresence of information technology and the vast array of digital opportunities have rapidly changed many aspects of our society and economy. It's only evident that digitalization, particularly connectivity and "smart" object technologies, is also expected to revolutionize industrial production, often paraphrased by the term "Industry 4.0". According to [8], it is an umbrella term for the following fundamental concepts:

- **Smart Factory:** With the aid of sensors, actors, and ubiquitous computing techniques, manufacturing plants are able to control their own production and run autonomously.
- **Cyber-physical Systems:** Physical and digital representation of production systems and products are so closely intertwined that they cannot be reasonably differentiated anymore.
- **Self-organization:** The decentralization of systems leads to a decomposed and self-organized production.
- **New systems in distribution and procurement:** Cross-organizational processes are highly connected and individualized.
- **New systems in the development of products and services:** Development processes are also individualized, making use of new innovation concepts.
- **Adaptation to human needs:** New systems are designed according to human needs, not vice versa.
- **Corporate Social Responsibility:** Manufacturing is increasingly influenced by sustainability and resource-efficiency.

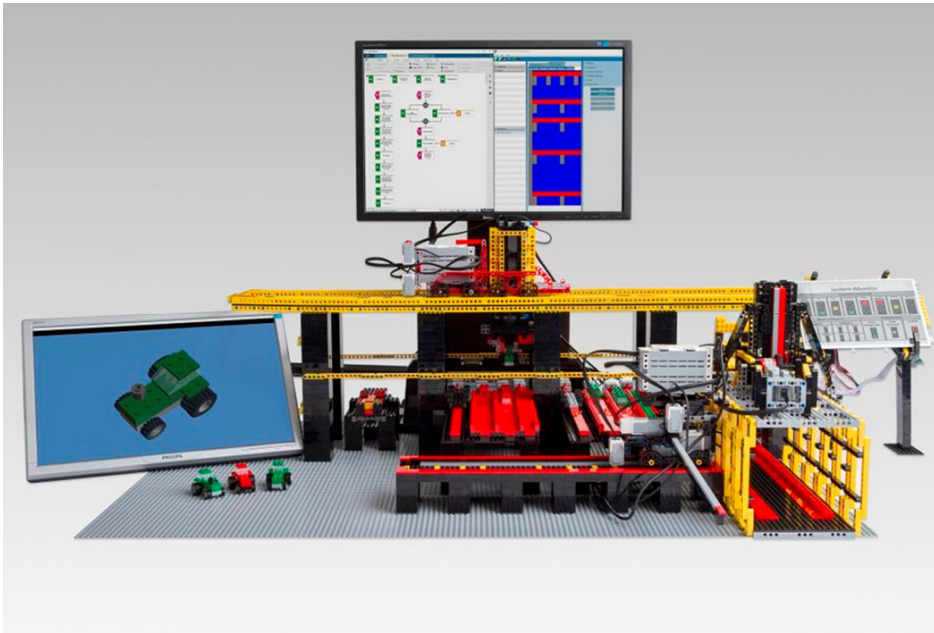


Figure 2: The DFKI-Smart-Lego-Factory and its production process.

The term “Industry 4.0” was originally coined in a research initiative by the German government. Often proclaimed as the fourth industrial revolution, the “version numbering” in the term “4.0” is a reference to software management. It should be noted that the English term “Industrial Internet” describes a closely related idea, however focusing even more on the application of IoT technologies in the industrial context.

3 The DFKI-Smart-Lego-Factory

To demonstrate the manifold potentials Industry 4.0 has for managing processes within and around smart manufacturing plants, the DFKI has constructed a factory out of LEGO® bricks, shown in Fig. 2. The DFKI-Smart-Lego-Factory manufactures different variants of LEGO® tractors in a fully automated way. Therefore, it combines the standard LEGO® bricks, more complex LEGO® Technic components, and the programmable LEGO® Mindstorms EV3 controllers and engines with the RefMod-Miner software,¹ a self-developed research prototype for the analysis of business process (models) and the development of reference models. The tractors are assembled from standard components built out of LEGO® bricks. Regarding the BPM lifecycle stages, the manufacturing process is defined as follows:

1. **Process Strategy:** The factory produces customized tractors to order. As usual in automobile production, multiple variants of the same standard product are offered. The lower-structure individual parts that can be freely combined into a total of 32 variants are kept in stock. Each product is individually manufactured on an autonomous production line.
2. **Process Modeling:** The factory is controlled by a flexible and configurable model of a production process in form of an Event-Driven Process Chain (EPC), which is embedded into a complete model of the value chain with regard to the concrete product. This model covers all possible production variants and is individually instantiated at the beginning of each production, representing the chosen product configuration.
3. **Process Implementation:** The factory is realized as a cyber-physical system (CPS), based on the semantics of the defined comprehensive process model. The individual production modules are associated with the production steps and controlled by software, digital equipment, and sensors alike.
4. **Process Execution:** The factory executes the production process autonomously, save for refilling the component supply with new parts. Therefore, it is equipped with color, contact, and infrared sensors, such that individual production modules can control themselves and communicate with each other. For instance, the crane uses color sensors to identify its current position and to determine the next actions to take.

¹ <https://refmod-mine.dfki.de>

5. Process Monitoring & Controlling: By permanently recording sensor data, the factory is also able to permanently monitor and control its current status and log the data of finished process instances.
6. Continuous Process Improvement: Using the collected process data, we are able to identify optimization potentials in the production process, such as suboptimally placed component supplies.

This one-of-a-kind factory was constructed from about 9,000 LEGO® bricks. It contains eight LEGO® Mindstorms EV3 controller modules, 14 individual motors, and a total of 23 sensors. The factory was in parts designed with the help of the LEGO® Digital Designer, an online Computer-Aided Design (CAD) tool, which allows to digitally model the construction before realizing it with actual bricks. It is controlled by the RefMod-Miner software, which calls the Java-based open source leJOS² framework running on the controller modules. A video of the factory and its production process can be found at <https://youtu.be/mNtZuZtg1dU>.

4 Application Case 1: Model-based management of manufacturing processes

4.1 Main concepts

Regarding the BPM lifecycle, Industry 4.0 can be interpreted as a new company strategy that translates into a new process strategy, with effects on both the product and the production [12]. Production systems are closely connected, with high modularization, horizontal integration across the value chain, and standardized interfaces for machine communications. Products provide their own ecosystem, with functionalities and interfaces for reacting to external input factors, collecting and exchanging data, and monitoring their own handling. A process strategy also has to account for producing low quantities of highly individualized products to meet specific customer needs, the management of autonomous transportation systems, and the monitoring of necessary collaborations between production machines and humans.

Leveraging these potentials requires drastically changing current manufacturing processes, going from a highly standardized, rigid workflow to a decentralized, highly flexible process. Since production modules commu-

nicate locally, it is difficult to assess the current status or the available information of the process or product. This is a challenge, because it must be possible to interfere with the running production (e. g., in case of a machine error or a change in order priorities), but also because collecting process execution data is important for making adjustments in later cycle stages. Business process modeling [10] in general and reference modeling [4] in particular are widely used BPM methods, which also offer potential to conquer challenges in Industry 4.0.

4.2 Application in Industry 4.0

Established BPM techniques like process modeling and reference modeling provide a promising approach for the management of smart factories. Implementing manufacturing processes in a smart factory with business process semantics combines central monitoring with decentralized self-organization. Each production step is modeled as a process function associated with the responsible production module, allowing changes to the production process without in-depth technical adaptations or knowledge. Innovative modeling approaches connect both modules and functions with, e. g., sensor objects or decision algorithms, which play a role in executing the respective function. At decision points, modules collaborate to find a mutual agreement on the process organization.

The uncommon application of BPM methods to manufacturing processes offers promising benefits for smart factories. Defining an autonomous manufacturing process by means of a process model introduces necessary dependencies, while maintaining a high degree of freedom for ad-hoc self-organization. Managing each production module separately facilitates decentralization and hence, flexibility. As each active production module reports execution data back to a monitoring unit, they leverage a central process instance control unit, which records data for future analysis. Finally, by modeling and monitoring the manufacturing process on a higher level of detail (including sensor and actor data), the entire factory can be overseen simultaneously, allowing for immediate reactions to mitigate process risks.

On a technical level, this means that each production module is a separate autonomous system, which is aware of its assigned functions and their position within the production process. Modules log the low-level data associated with their production step on the shop-floor and communicate directly with each other via technical interfaces. They report their current status and the collected data points to a central process instance control unit, which acts as

² <http://www.lejos.org/>

an additional data integration and process management layer. By aggregating the data from all production steps, the control unit is able to monitor the currently running process instances and assess the overall realtime status of the smart factory. Information on the process performance are gained by evaluating the stored data from past instances.

In order to leverage these potentials for newly planned smart factories, it might also be beneficial to extend established BPM techniques such as the EPC with manufacturing-specific or Industry-4.0-specific concepts, which allow for a better understanding and more precise management of the manufacturing process. Such a domain-specific modeling language for digitized manufacturing processes could include details and constraints in the execution that domain-independent modeling languages are unable to capture, such as employee safety regulations or machine-to-machine communication channels. A precise digital model of manufacturing-related concepts would enable an even closer interrelation between the tangible and digital process representations.

4.3 Implementation in the DFKI-Smart-Lego-Factory

In the example of the DFKI-Smart-Lego-Factory, we implemented a manufacturing process for the individual production of tractors, with autonomous transportation systems moving the individual parts and the finished product through the separate production modules. The process starts once a new individual production order arrives after being entered in the order form. As each ordered “product” is customized to the buyer’s preferences (“batch size 1”), make-to-stock production is infeasible. However, a single tractor is made out of several standard components, so the component supply has to be constantly re-stocked. The production process, by which the factory is managed, is shown on a status monitor in the background (see Fig. 2). During production, the current function is highlighted in yellow (see Fig. 3).

Besides classical process modeling, other graphical visualization techniques are necessary to monitor the production environment, which is closely linked with the production process, sensors, and actuators in the smart factory. This is why our factory includes the control panel, shown in Fig. 4. Rectangles represent production modules, with colors indicating a module’s current status. A red module indicates a faulty status and requires an intervention by the factory supervisor. Traffic lights represent sensors, with the different light colors having different seman-

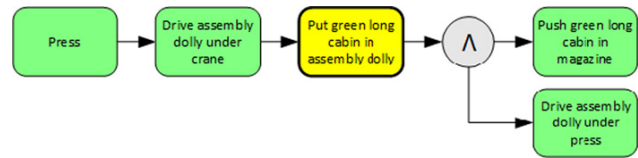


Figure 3: Highlighting the current process step.

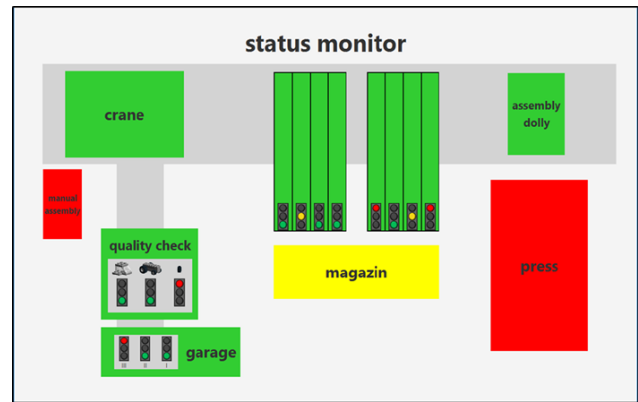


Figure 4: The status monitor.

tics for each sensor type. For the component supply in the center, a red light means that the stock is empty and has to be refilled before the next process instance. In quality control, a green light means that the tractor complies with the quality regulations (colors and exhaust assembled according to order specification). A red light in a garage means that this compartment is already filled with a finished tractor, which has to be removed before a new one is delivered. The status monitor updates immediately once the sensor measurement recognizes a status change.

Both discrete process step highlighting and the status monitor are domain-specific extensions of the standard EPC, which is used to model and execute the manufacturing process. There is further potential for the development of integrated manufacturing modeling concepts, which would allow for depicting the direct link between the process, its sensors and actuators, the currently running instance, and the smart factory as a whole.

5 Application Case 2: Process mining for manufacturing processes

5.1 Main concepts

Following the BPM Lifecycle, process monitoring is followed by continuous process improvement. Efficient and

effective processes contribute to the economic efficiency of the organization, therefore process improvements are always of interest. Established data mining and machine learning techniques offer a data-driven approach to process optimization [11]. Process mining combines methods from data mining with business process management domain knowledge to analyze collected process data. By extracting knowledge from process logs, these techniques allow stakeholders to gain new insights into their processes, such that their behavior can be continually improved.

Standard process mining approaches work on logs produced by workflow-supporting process execution systems. We distinguish three major areas, focusing either on discovering as-is process models from instance-level log data, checking the conformance of process logs with a predefined process model, or enhancing a given process model to optimize the underlying process [1]. Process data is typically collected on event-level, denoting the execution of a certain activity within a process, along with a timestamp. Additionally, resources, such as employees or machines, or other relevant data can be attached to both process events and process instances, enabling the comparison of different process variants.

5.2 Application in Industry 4.0

So far, process mining is not frequently applied in manufacturing. This is mainly due to not having access to the data required for data-centered process analytics, currently one of the main challenges in production process optimization. The technical opportunities of Industry 4.0 now allow us to collect process execution data with a much higher frequency and level of detail, enabling us to exploit previously untapped potentials in the analysis of production processes [7]. By constantly logging the process data that is gathered by the software, sensors, and cyber-physical components within a smart factory, we are able to monitor the status of both the process and the production system, enabling a real-time controlling of the running business process instances during production.

Closely linking the manufacturing process with its digital representation can simplify its management [7]. On one hand, IoT systems and sensors may automatically track executed process activities on a much lower-level

and mitigate the dependency on wrongly or sparingly documented process data. On the other hand, integrating collected data across all production stages, including manufacturers and customers allows us to inspect both the process and the product in a holistic way. With all the available data, smart factories contribute to paving the way for leveraging benefits of process mining in actual manufacturing contexts in order to identify concrete potentials for process optimization.

5.3 Implementation in the DFKI-Smart-Lego-Factory

The DFKI-Smart-Lego-Factory extends process mining analysis to the shopfloor to gain a deeper understanding of production process behavior. During the CeBIT 2016 exhibition, we collected execution data for a total of 309 tractor production process instances to use for process mining, with first results shown in Tab. 1. We can see that the maximum processing time of a process instance was about 17 minutes. After analyzing the instance in question, it was found that this process instance has remained in the “show assembly video” process step for a very long time, indicating a long delay in the built-in manual assembly step. This resulted in a delayed overall process and ultimately a high deviation from the average process duration. The high standard deviation can also be explained by this process step. Whereas the automated manufacturing process steps have a fairly standardized runtime (with some deviation caused by the mechanical parts), the execution of the manual step depends on whether an employee is available for its execution. If this is not the case (for example, because the employee is busy talking to interested exhibition visitors), the factory waits longer for the process instance to be completed.

Furthermore, process mining can identify deviations in the production process based on the manufactured variations. For example, the factory produces tractors of two different colors (red and green). Table 2 shows the execution time statistics. We observe that producing a green tractor is generally faster than producing a red tractor. This is due to the physical structure of the system, where the components supply is located in the middle of the factory. Red parts are located further away from the assem-

Table 1: Process Execution Time Analysis.

	Minimum	Average	Median	Maximum	Std. Derivation
Process Lead Time	2 m 47 s	4 m 24 s	3 m 14 s	17 m 24 s	1 m 57 s

Table 2: Process Variant Comparison.

Function	Model 1 (green)		Model 2 (red)	
	Average	Std. Deviation	Average	Std. Deviation
put chassis in assembly dolly	15.9 s	14.8 s	13.1 s	3.4 s
put body in assembly dolly	22.3 s	9.1 s	25.2 s	5.7 s
put long cabin in assembly dolly	18.4 s	1.0 s	22.2 s	1.7 s
put short cabin in assembly dolly	18.8 s	0.9 s	22.7 s	0.8 s

bly dolly than green parts, such that the crane has to cover a wider distance transporting red tractor parts to the assembly step. This results in higher execution times for red tractors than green ones. If a factory were to produce more red tractors, it would make sense to switch the slots of red and green parts or rearrange the component supply as a whole.

6 Application Case 3: Deep-learning-based prediction for manufacturing processes

6.1 Main concepts

Model-based process management is an ex-ante approach to define and control an autonomous production. Process mining on production data analyzes completed process instances ex-post to determine their performance. It is equally interesting and potentially more helpful to manage and control running process instances during their execution, such that abnormal process behavior is detected before problems arise [6]. Real-time business process management allows, e.g., case managers to intervene early if processes are likely to be late or terminate abnormally. Intervening early to focus on resolving potential issues may mitigate business risks and save resources [2]. Process prediction is an important part of managing busi-

ness processes at runtime. It aims at predicting information about future process behavior from past process execution data, i. e., logs containing information on, e.g., individual case data, event sequences, or environmental factors as recorded by process execution systems (cf. Fig. 5). Approaches exist for predicting the overall outcome of a running process instance, process performance indicators (PPIs) such as throughput times, or the next occurring process events [3].

6.2 Application in Industry 4.0

For the process execution data available in an Industry 4.0 scenario, we have derived a process prediction approach based on deep learning [3]. Process execution logs are encoded into a so-called embedding space and used as input for a recurrent neural network (RNN) with two hidden layers of long short term memory (LSTM) cells. Over the course of many training epochs, the network learns the relations between the individual process events as well as the associated organizational resources and execution times, implicitly representing the process semantics through its internal configuration.

When confronted with an incomplete process instance, the trained neural network is able to predict the next events and the associated resources. It can also estimate the remaining time required for completing the current process step as well as the entire process instance.

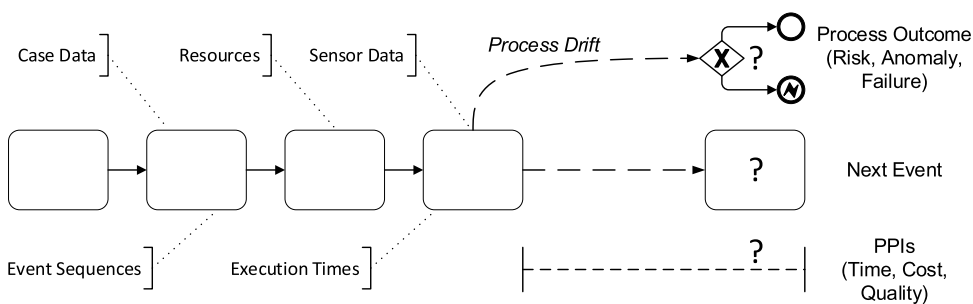


Figure 5: Input data and predictands of process prediction.

This is interesting for a manufacturing process, as potentially unknown relations between input factors can be discovered and factored into the prediction. Also, production can be planned and optimized in realtime, as the predicted time to completion is newly calculated after each completed process step. Since more reliable information on the process instance is available, the prediction becomes more accurate the less time remains until its completion.

6.3 Implementation in the DFKI-Smart-Lego-Factory

In the DFKI-Smart-Lego-Factory, we have included a trained neural network to predict next events and remaining time to completion of the current process instance. A special graphical user interface (GUI), shown in Fig. 6, shows the ever-adapting prediction results. It is separated in three panels. The left one shows the currently executed process step and the time remaining until its completion. The process step predicted to be executed next is shown in the center, and on the right, we see the estimated time required to finish the current process instance (right). The orange status bar in the left panel shows the progress of the current process step in relation to the predicted overall duration, whereas the status bar in the right panel shows the progress of the already completed process steps as compared to the predicted time to completion.

When trained on the tractor manufacturing process, the network learns for example that a green tractor body will most likely be followed by a green cabin cell, that visitors prefer long cabin cells over short ones, and that constructing a red tractor takes longer to complete than a

green one. All these factors have an influence on the predicted series of events as well as the estimated times of completion.

We measured the performance of the next-event process prediction by evaluating the instance data set collected during CeBIT 2016. The data was randomly split into a validation set and a test set, with a relation of 80 % to 20 %. The RNN was trained on the validation set; the trained net was then used to predict the instances from the test set and determine their correctness. The prediction achieved a precision value of 92.12 %, a recall value of 93.34 %, and an f-measure of 92.6 %.

In this scenario, neither the process model nor the instance configuration data is used for the prediction, such that the neural network not only learns the process structure and typical configurations from data on process execution, but also the associated execution times. This process step prediction is particularly useful, if no pre-defined process model is known.

7 Conclusions

In general, our contribution shows that essential aspects and challenges from Industry 4.0 can be addressed by suitable BPM methods. The DFKI-Smart-Lego-Factory acts as a learning and communication vehicle for future concepts and implementation projects, as it covers many real-life challenges. While the three cases we show here are all of independent interest, they are far from exhausting the demonstration potential of the DFKI-Smart-Lego-Factory or the application potentials of BPM for Industry 4.0. Other potential cases include:

- Plug-and-play production modules: Pre-configured production modules to be automatically integrated into a production process
- Predictive Maintenance: Monitor motor and sensor data to predict errors or failures
- MES Connection: Apply data analytics and machine learning to detect patterns in machine usage
- Data-driven Business Models: Use process mining results for optimizations, e. g., shop-floor redesign
- Process Configuration Recommendation: Recommend an optimal instantiation on a shop-floor with configurable production islands
- Complaint Management: Collect and analyze production data to predict and avoid customer complaints
- Inductive Reference Modeling: Collect data across multiple smart factories to find commonalities and “best practices” [9]



Figure 6: The prediction GUI.

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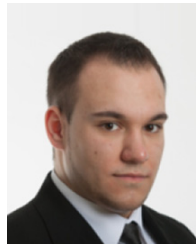
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